

The role of cyberbystanders: Agent-based models of bystander impact on online abuse

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Abstract—Digital communication has enabled harmful behaviours such as bullying to migrate rapidly online, disproportionately affecting children, women, and minority groups. Effective cyberbullying interventions require understanding the number, type, and roles of participants in abusive events. This study uses an agent-based model (using NetLogo) to simulate cyberbystander behaviour and assess how their actions influence the severity of abuse toward a victim. The model is parameterised with a publicly available cyberbullying dataset containing labelled participant roles and abuse levels. NetLogo simulations show that neutral bystanders—those who neither support nor oppose the abuser—contribute to escalating abuse, as their silence may be perceived as endorsement. In contrast, defenders who support the victim significantly reduce abuse, with both their numbers and early intervention playing a critical role. These findings highlight the importance of bystander-education programs that promote proactive and timely upstanding behaviour when witnessing online abuse.

Index Terms—Cyberbullying, Bystanders, Agent-based models

I. INTRODUCTION

The National Bullying Prevention Centre [1] defines cyberbullying as “unwanted, repeated, aggressive, negative behaviour that takes place over digital devices, and can occur through texting, on social media, during gaming, on instant messaging, and via photo sharing.” The eSafety Commissioner reported a 40% increase in

reported cyberbullying incidents in 2025 [2], with over 60% of these reports involving children aged 12-15. Cyberbullying incidents involve three main participant groups - *perpetrators*, who initiates the bullying; the *victims*, who are the target and recipient of harmful media (text, audio, video, images); and the *bystanders*, who witnesses the abuse [3]. Notably, bystanders are not merely passive observers; their actions—such as liking, sharing, or participating—can amplify harm, whereas intervention can uphold social norms and mitigate negative outcomes.

Given the complexity and interconnectedness of social interactions in cyberbullying, robust tools are required for meaningful analysis. Agent-based models (ABMs) are increasingly leveraged within computational social science for this purpose, as they enable researchers to simulate autonomous individuals (agents) whose rule-based interactions reveal emergent group- or system-level outcomes [4]. Despite advances in the application of ABMs to social behaviour, the intersection of online abuse, bystander intervention, and uncertainty remains inadequately investigated, with only limited agent-based simulation of offline bystander dynamics [5].

To address this gap, we develop an ABM tailored to cyberbullying contexts, simulating individual decision-making and its impact on the broader online envi-

ronment. We train the model on the CYBY23 cyberbullying dataset [6], which provides labelled roles for participants in each session, allowing for a data-driven exploration of bystander intervention dynamics.

In the remainder of this paper, Section II reviews literature on modelling bystander behaviour in cyberbullying events. Section III outlines our methodology and dataset. Section IV presents our findings and analysis. Finally, Section V provides some discussions, future research directions, and conclusions.

II. RELATED WORKS

Kowalski et al [7]’s meta-analysis found that cyberbullying perpetration correlated with normative beliefs about aggression and moral disengagement, while victimisation was closely related to elevated stress and suicidal ideation. Recent research has shown that bystander behaviour is driven by complex dynamics impacted by empathy, moral disengagement, and group norms.

A foundational concept in bystander behaviour is the bystander effect, introduced by Latané and Darley [8] following the murder of Kitty Genovese [9]. This theory suggests that individuals are less likely to intervene when others are present, due to a complex decision-making process rather than mere apathy. In digital spaces, the anonymity and scalability of online platforms intensify these dilemmas, intervening in cyberbullying scenarios even less likely. Shen et al. [10] confirmed that these factors predict whether individuals defend, ignore, or exacerbate online aggression. Barlett et al. [11] showed that cultural differences, such as collectivist values and trust in institutions, shape tendencies to intervene. These insights show that bystander behaviour in cyberbullying contexts is highly flexible and shaped by a range of individual, social, and platform-specific influences.

Given this dynamic and multifaceted nature, ABMs have proven effective in investigating bullying processes due to their capacity to simulate heterogeneous agents and complex social interactions. Xanthopoulou et al. [12] introduced the MARG model, viewing bullying as a manifestation of intensified marginalisation emerging from everyday social dynamics, rather than predefined roles. Their norm-driven, preference-based

partner selection framework demonstrates how exclusionary behaviours and bullying can arise organically within university student populations. Abraham [13] explored bullying as part of broader social development in education, representing individuals as adaptive agents whose behaviour evolves through repeated interactions and feedback. This work shows how patterns of aggression and victimisation emerge or dissipate based on interaction dynamics, underscoring the developmental and systemic nature of bullying. Ide [14] developed an ABM to analyse intervention strategies in suppressing bullying, focusing on agent behavioural traits and social feedback. The study found that altering individual behaviours and responses can substantially affect bullying dynamics, emphasising the applicability of ABMs for evaluating prevention and intervention strategies within social systems.

ABMs have also gained traction in the study of cyberbullying, offering a powerful approach to simulate the complex interactions in online social networks. Recent work [15] introduced a data-driven ABM for public opinion propagation simulation to analyse the spread of public opinion during cyberbullying incidents on social media. Their model integrates opinion dynamics and epidemic modelling principles, employing a fine-grained ABM within a cyber-physical-social space to simulate how user interactions, opinion fusion and polarisation, local popularity, and topic interest drive the evolution of online public opinion. In [16], the detection of cyberbullying in digital media, particularly memes, using ABMs has been explored, and the shortcomings of traditional captioning methods and limited dataset scopes were addressed. A multi-modal ABM has been developed, where large multimodal models (LMMs) have been employed as specialised agents and LMMs as vision experts. These agents collaboratively analysed images, posed insightful questions tailored to identifying potential social abuse, and iteratively consulted the LMM for enriched visual descriptions. Their findings offered a robust, comprehensive meme analysis and advanced the detection of harmful content across online platforms.

However, to date, there are no published studies applying ABM specifically to bystander behaviours in the cyberbullying context and this work is a significant

first step in that direction.

III. METHODOLOGY

In this section, we describe the dataset utilised in this study and outline the ABMs applied to analyse cyberbullying incidents within this data.

A. Dataset

The Cyberbullying Bystander Dataset (CYBY23) [17], was developed by researchers at the Universiti Sains Malaysia and Qassim University [6]. Collected in real-time via the Twitter API from January 2022 to January 2023, the dataset focused on content likely to be aggressive or controversial, using keywords and hashtags linked to sensitive topics such as religion, ethnicity, immigration, sarcasm, and racial bias. The initial dataset comprised 150 original discussions; following a multi-stage annotation process, 112 threads (639 tweets) were selected for final annotation and validation. Each thread begins with a provocative or aggressive initiating comment, followed by responses capturing diverse bystander behaviours.

Annotation involved three stages: (1) tweet-level classification for aggression (0 = low, 1 = moderate, 2 = high); (2) response-level classification: *Instigators* (agree), *Defenders* (disagree), *Neutral* (impartial bystanders), or *Others* (unrelated); and (3) thread-level verification, assessing the context of entire conversations. Five annotators from diverse backgrounds independently coded each instance using clear role definitions, and majority voting was applied for reliability. Fleiss’s Kappa coefficient ($\kappa = 0.6078$) indicates substantial agreement. Table I summarises the variables for each tweet.

1) Data Analysis

To characterise the dataset, two main variable groups were analysed: content-based aggression variables (*Identity_attack*, *Profanity*, and *Toxicity*) and engagement variables (*Retweet_count* and *Favorite_count*).

As shown in Fig. 1, all content-based variables exhibited strong right skewness, with most values concentrated near zero and a minority forming long tails towards higher values. *Identity_attack* showed the most severe skew, with over 80% of values below 0.1, highlighting both the infrequency of overt identity-based hostility and the presence of sporadic

TABLE I: Summary of variables in CYBY23

Variable name	Type	Description
User	String	Username of the Twitter account that published the message
User_id	Numeric (String ID)	Distinct Twitter user identifier allocated by the platform
Tweet_id	Numeric (String ID)	Unique identifier for the tweet
Reply_id	Numeric (String ID)	Identifier shared by all tweets in the same thread
Created_at	Date/Time	UTC timestamp for tweet creation
Retweet_count	Integer	Number of retweets at time of collection
Favorite_count	Integer	Number of likes at time of collection
Text	String	Complete tweet text including punctuation, emojis, and hashtags
Media_url	String	Link to attached media if present
<i>Features Captured using Perspective API</i>		
Insult	Binary (0/1)	Automated flag for offensive language
Threat	Binary (0/1)	Detection of explicit threat terminology
Identity_attack	Binary (0/1)	Identity-based attacks (race, gender, religion)
Profanity	Binary (0/1)	Use of explicit or vulgar language
Toxicity	Float (0–1)	Probability score of overall toxicity
Severe_toxicity	Float (0–1)	Probability of high-severity harmful content
<i>Features Captured using Textblob</i>		
Polarity	Float (0–1)	Lexical mood polarity
Subjectivity	Float (0–1)	Degree of opinionated language
Sentiment	String	Polarity-based sentiment class
<i>Features Captured using Human Annotators</i>		
Class_label	Categorical (0–2)	Thread Label 0 = none, 1 = low bullying, 2 = high bullying
Bystander_roles_label	Categorical (0–3)	0 = Instigator, 1 = Defender, 2 = Impartial, 3 = Other

cases of extreme abuse. *Profanity* demonstrated a similar distribution, though with a slightly broader spread and occasional spikes in explicit content. In contrast, *Toxicity* was more evenly distributed across moderate values, indicating a greater prevalence of mild hostility within online conversations.

As illustrated in Fig. 2, the engagement metrics exhibited even greater asymmetry. Both *Retweet_count* and *Favorite_count* demonstrated long-tailed, zero-inflated distributions typical of web engagement data, where the majority of posts receive minimal or no attention [18]. While 95% of tweets were retweeted fewer than ten times, a small minority exceeded several thousand retweets. Similar trends were observed for *Favorite_count*, with most posts receiving zero interactions, but about 1% reaching into the tens of thousands. These results underscore the strong social amplification effect present in digital communication, highlighting the importance of visibility in the spread of online aggression.

Tables II and III present summary statistics

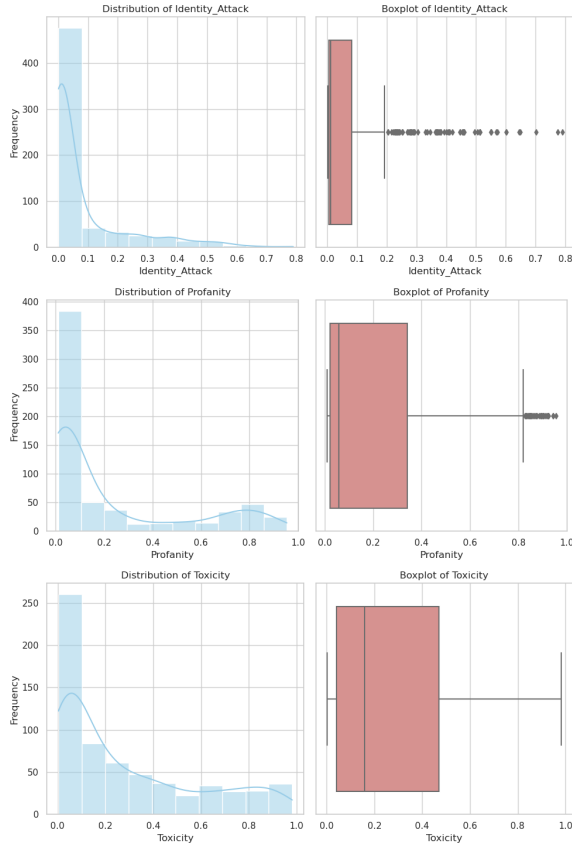


Fig. 1: Empirical distributions and boxplots of content-based aggression variables (Identity_attack, Profanity, Toxicity).

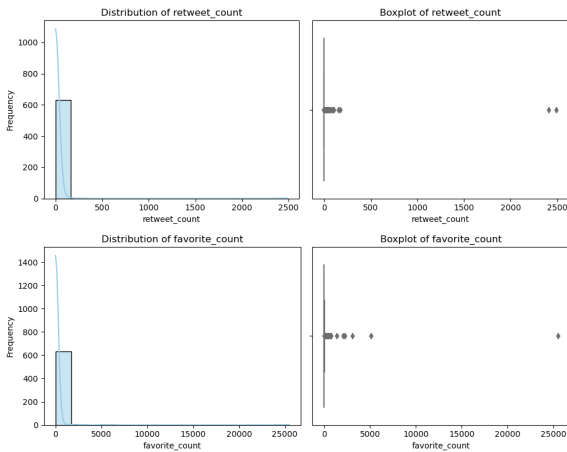


Fig. 2: Empirical distributions and boxplots of engagement variables (Retweet_count, Favourite_count).

and quartile measures for the key behavioural variables: Retweet_count, Favourite_count, Toxicity, Profanity, and Identity_attack. The results confirm the right-skewed nature of the dataset, with median Retweet_count and Favourite_count near zero, yet maximum

values reaching 2,490 and 25,476, respectively. This distribution reflects common social media dynamics in which the majority of posts attract minimal attention, while a minority attain exceptionally high visibility.

TABLE II: Descriptive statistics

Variable	Count	Mean	Med.	Mode	SD	Min	Max	Skew	Kurt
Retweet_count	636	10.23	0.00	0.00	137.79	0.00	2490	17.55	309.81
Favourite_count	636	79.28	0.00	0.00	1046.69	0.00	25476	22.81	549.01
Identity_attack	636	0.08	0.01	0.10	0.14	0.00	0.79	2.23	4.57
Profanity	636	0.23	0.06	0.71	0.30	0.01	0.95	1.23	-0.09
Toxicity	636	0.28	0.16	0.38	0.29	0.00	0.98	0.95	-0.44

TABLE III: Quartile summary of variables

Variable	Q1 (25%)	Q2 (Median)	Q3 (75%)	Q4 (Max)
Retweet_count	0.00	0.00	0.00	2490.00
Favourite_count	0.00	1.00	3.00	25476.00
Identity_attack	0.00	0.01	0.08	0.79
Profanity	0.02	0.06	0.34	0.95
Toxicity	0.04	0.16	0.47	0.98

Toxicity, Profanity, and Identity attack also exhibited pronounced skewness, with mean values of 0.28, 0.23, and 0.08, respectively, and maximum values approaching 1. These results suggest that while most interactions were mild, a small number of highly toxic posts had a substantial impact on the overall environment. The skewness and kurtosis values further reinforced this pattern, particularly for Retweet_count (skewness = 17.55, kurtosis = 309.81) and Favourite_count (skewness = 22.88, kurtosis = 549.01), both of which demonstrated extreme overdispersion.

TABLE IV: Summary of bystander roles per session and across CYBY23

Bystander roles	Mean proportion per session	Median proportion per session	Proportion across all sessions
<i>Instigator</i>	0.18	0.00	0.44
<i>Defender</i>	0.31	0.24	0.18
<i>Neutral</i>	0.50	0.50	0.36
<i>Other</i>	0.01	0.00	0.02

2) Bystander Analysis

Bystander roles labelled in CYBY23 and described in Table I are *Instigators*, *Defenders*, *Neutral*, and *Others*. Fig. 3 illustrates the distribution of these roles across sessions (threads), with a maximum of 11 bystanders per session. *Instigator* and *Defender* distributions are heavily skewed; most sessions have none or one *Instigator/Defender*, though a minority have up to 10 *Instigators* or 8 *Defenders*. Table IV indicates that, on average, 30% of bystanders are *Defenders*, supporting previous findings that intervention rates decrease with multiple bystanders present [19], [20].

Fig. 4 displays participant distribution by aggression levels, revealing that sessions with high bullying (label = 2) feature more *Instigator* and *Neutral* bystanders, but fewer *Defenders*. This pattern suggests that *Defender* responses remain rare and an abundance of *Neutral* bystanders may contribute to heightened cyberbullying. Results highlight the critical role of *Defender* in mitigating the severity of bullying incidents.

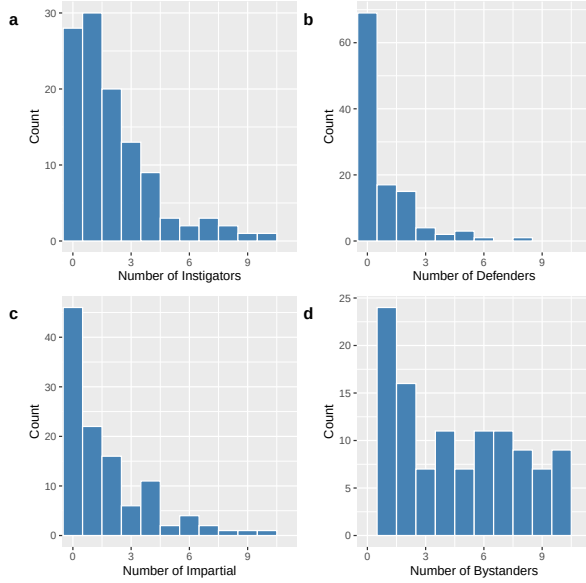


Fig. 3: Distribution of bystander roles per session

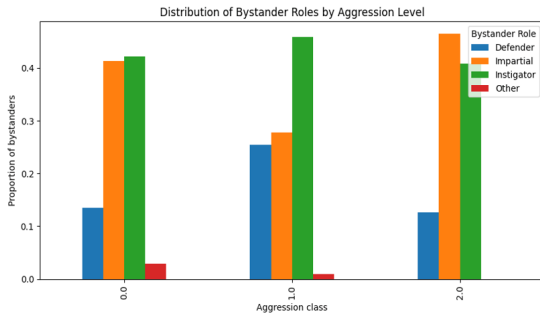


Fig. 4: Proportions of bystander roles for three levels of bullying: 0 indicates no bullying, 1 indicates low bullying and 2 indicates high bullying

B. Agent-based Models for cyberbullying incidents

Cyberbullying incidents were simulated using NetLogo¹, with system parameters informed by observations from the CYBY23 dataset.

1) Model overview

Each simulation comprises one perpetrator (abuser), one victim, and multiple bystanders, each assigned

¹The NetLogo code is available at <https://github.com/hrishika1208/Bystander-behavior-ABM/blob/main/NetLogo%20code>

one of four roles: *Instigator*, *Defender*, *Neutral*, or *Other*. The abuser's actions are modelled as movement toward the victim, with environmental variables—including Toxicity, Profanity, Identity_attack, Retweet_count, and Favourite_count—determining both the initial aggression level and its escalation. When the abuser's aggression reaches a threshold, bystanders may intervene based on their roles: *Defenders* move toward the victim to reduce aggression; *Instigators* move toward the victim and increase aggression; *Neutral* and *Other* bystanders have no impact. The simulation terminates when aggression either hits a maximum value or falls below a minimum threshold due to *Defender* intervention, the latter indicating a retreat by the perpetrator. Fig. 5 depicts the agent workflow in the NetLogo model.

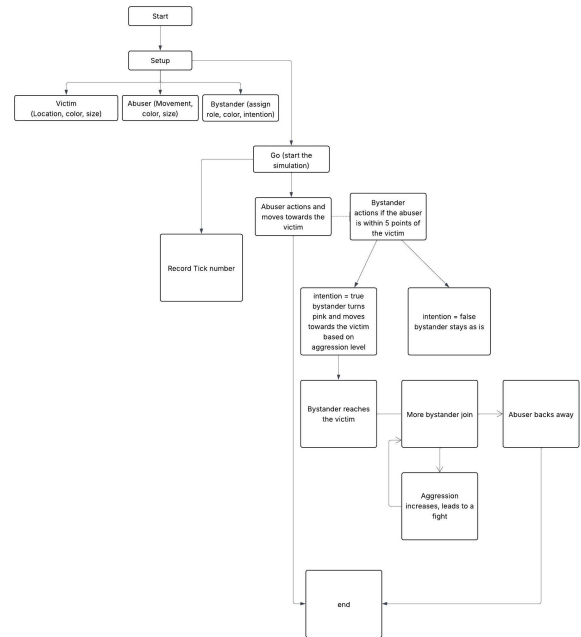


Fig. 5: Agent workflow in the NetLogo model

2) Model parameters

Model parameters were selected to closely replicate the dynamics observed in the CYBY23 dataset. Key parameters include:

- *Number of bystanders*. Randomly drawn between 5 and 15, consistent with Fig. 3;
- *Aggression increase factor*. Defined as a linear combination of five environment variables: Toxicity, Profanity, Identity_attack, Retweet_count, and Favourite_count, each weighted to reflect their causal influence. This factor

is computed as follows:

$$\begin{aligned} \delta F = & W_t \times \text{Toxicity} + W_i \times \text{Identity_attack} \\ & + W_p \times \text{Profanity} + W_r \times \text{Retweet_count} \\ & + W_f \times \text{Favorite_count}. \end{aligned}$$

To model the observed heavy-tailed distributions (see Figs.1,2), variables were sampled from Pareto distributions, with the values of α given in Table V;

- *Initial aggression level.* Calculated as $30 + (10 \times \delta F)$, simulating environmental impacts;
- *Bystander reaction trigger.* Bystanders act when the abuser approaches within 5 steps of the victim, reflecting a minimum aggression threshold for intervention;
- *Bystander role choice.* Triggered bystanders choose a role; *Defenders* appear 30% of the time, matching empirical session averages (Table IV);
- *Abuser retreat threshold.* *Defender* actions decrease aggression by 10 units; *Instigator* actions increase it by 5 units. If *Defender* actions reduce aggression below 20% of its original value, the abuser retreats, ending the simulation;
- *Maximum aggression threshold.* If aggression reaches 100 units, the simulation ends, denoting successful bullying escalation;
- *Maximum simulation time.* Simulations terminate after 200 time steps if no stopping conditions are met.

TABLE V: Parameters to set global aggression

Metric	Derivation / Tuning Basis	Alpha (α)	Weight W_s
Favorite_count	Derived from Q3 = 3	1.26	$\frac{1}{40}$
Retweet_count	Tuned for extreme skew and stability	1.50	$\frac{1}{20}$
Toxicity	Tuned to reflect high relative skew	1.80	2
Profanity	Tuned for moderate skew	2.20	1
Identity_attack	Tuned for the lowest skew and highest stability	2.50	1.5

Fig. 6 presents a simulation of the NetLogo model, illustrating the variables influencing bystander behaviour. In this example, the simulation features two *Defenders*, two *Neutrals*, and one non-volunteer, which leads to a rapid decrease in aggression.

IV. EXPERIMENTS

We have conducted 40 simulations. In each simulation, the number of bystanders, type of bystander (*Instigator*, *Defender*, *Neutral*, and *Other*), Aggression change, Tick numbers, and Stopping condition were recorded. Table VI captures details of all 40 runs of the model.

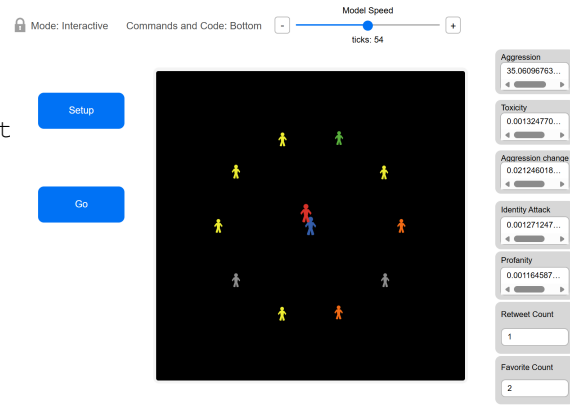


Fig. 6: Simulation of the NetLogo model.

Out of the simulations run, 17 of them stopped when the aggression dropped below 20% of initial values and the abuser retreated. In more than 80% of these scenarios, the number of *Defenders* was equal to or more than the number of *Instigators*. In 3 of the simulations, all bystanders chose to act, but their collective action caused no change in the aggression level. In all other 20 simulations, the aggression level neither dropped by 20% of the initial value, nor reached the maximum limit set for the environment, and stopped once it timed out. In more than 50% of these simulations, the number of *Instigators* was more than the *Defenders*. These simulations represent scenarios where *Defenders* are not successful in stopping abuse and moderate levels of bullying continue to remain in the session.

On simulations 1 and 37 (Table VI), we observed that *Defenders* were present and acted early causing environment aggression level to increase slowly or decline quickly. In addition, in simulations 5, 12, and 18, *Defenders* outnumbered the *Instigators*, aggression decreased rapidly. In Simulations 10, 14, and 28, we observed that the aggression level first became negative causing the abuser to back away, but when more *Instigators* joined later, the abuser gained support and started moving towards the victim again causing increase in aggression. In simulations 23 and 27 as well, the *Defenders* came in early and *Instigators* joined much later and although the aggression increase occurred, it still wasn't enough to counteract the *Defenders* effect, causing the abuser to back off. In contrast, environments characterized by larger number of *Instigators* demonstrated rapid escalation. This pattern aligns with findings from Gerritsen [5], who applied

TABLE VI: Summary of simulation results (I/D/N/O indicates *Instigators*, *Defenders*, *Neutral*, and *Others*)

Simulation	Bystander	I/D/N/O	Aggregation	Ticks	Stopping Condition
1	13	2/4/4/3	0.75	10	Aggressor backed away due to bystander action
2	6	2/2/1/1	0.80	16	Aggressor backed away due to bystander action
3	9	2/3/3/1	0.10	7	Aggressor backed away due to bystander action
4	14	3/1/1/1	0.31	200	Number of ticks reached 200 and the simulation stopped
5	16	0/2/1/2	-0.06	37	Abuser backed away due to bystander action
6	13	3/5/2/3	0.14	200	Number of ticks reached 200 and the simulation stopped
7	12	2/3/4/2	0.11	200	Number of ticks reached 200 and the simulation stopped
8	8	4/1/2/1	0.30	200	Number of ticks reached 200 and the simulation stopped
9	17	1/3/1/7	0.08	200	Number of ticks reached 200 and the simulation stopped
10	12	1/3/2/3	0.05	200	Number of ticks reached 200 and the simulation stopped
11	16	5/3/4/4	0.26	200	Number of ticks reached 200 and the simulation stopped
12	12	2/5/2/3	-0.10	70	Abuser backed away due to bystander action
13	15	1/1/2/2	0.08	200	Number of ticks reached 200 and the simulation stopped
14	15	5/4/3/3	0.00	200	Number of ticks reached 200 and the simulation stopped
15	17	5/4/4/3	-0.05	200	Abuser backed away due to bystander action
16	12	1/1/2/0	0.10	200	Number of ticks reached 200 and the simulation stopped
17	14	4/5/0/3	0.80	200	Number of ticks reached 200 and the simulation stopped
18	14	1/3/7/3	-0.08	72	Abuser backed away due to bystander action
19	8	2/0/3/2	0.14	200	Number of ticks reached 200 and the simulation stopped
20	12	3/5/3/1	0.25	200	Number of ticks reached 200 and the simulation stopped
21	13	0/2/2/2	-0.02	200	Number of ticks reached 200 and the simulation stopped
22	14	6/3/4/1	-0.06	70	Abuser backed away due to bystander action
23	11	2/2/6/1	-0.02	121	Abuser backed away due to bystander action
24	8	5/0/1/2	0.12	200	Number of ticks reached 200 and the simulation stopped
25	13	1/3/6/3	-0.02	100	Abuser backed away due to bystander action
26	9	2/1/3/3	0.07	200	Number of ticks reached 200 and the simulation stopped
27	9	1/4/3/1	-0.01	108	Abuser backed away due to bystander action
28	15	4/3/3/5	0.01	200	Abuser initially backs away but later re-approaches with reduced aggression
29	15	3/8/1/3	-0.04	119	Abuser backed away due to bystander action
30	14	7/2/3/2	0.11	200	Number of ticks reached 200 and the simulation stopped
31	12	6/5/0/1	-0.03	150	Abuser backed away due to bystander action
32	10	1/3/2/4	-0.07	100	Abuser backed away due to bystander action
33	8	1/2/2/4	0.00	120	All bystanders acted up and no change in aggression recorded
34	13	3/7/1/2	-0.05	192	Abuser backed away due to bystander action
35	14	7/3/2/2	0.07	200	Number of ticks reached 200 and the simulation stopped
36	12	3/5/2/2	-0.04	145	Abuser backed away due to bystander action
37	12	2/4/5/1	-0.06	146	Abuser backed away due to bystander action
38	8	2/2/1/3	0.05	120	All bystanders acted up and no change in aggression recorded
39	11	1/2/3/5	0.01	77	All bystanders acted up and no change in aggression recorded
40	10	2/1/3/4	0.04	110	All bystanders acted up and no change in aggression recorded

the bystander-effect framework to an agent-based context and showed that the timing and likelihood of intervention critically influence escalation dynamics. Our experiments show that aggression dynamics are path-dependent and that early hostility strongly affects the likelihood of recovery.

V. DISCUSSION AND CONCLUSION

The agent-based model (ABM) served as a simulation sandbox to examine how bystanders respond to different online environments and moderation conditions before real-world implementation. It captured emergent properties of collective aggression, demonstrating that de-escalation depends not only on individual ethics but also on network composition and the temporal sequencing of interventions. The model therefore provides a link between micro-level decision rules (e.g., *Defenders*' probability of acting) and macro-level aggression trajectories. The model also helps identify which behavioral factors, such as active intervention, passive observation, or social amplification,

most strongly determine whether an incident escalates or de-escalates.

Findings from the simulation study indicate that bystander roles have a substantial influence on mitigating aggression in online interactions. We clearly observed that *Neutral* bystanders who do not take action against an abuser can lead to high levels of bullying against a victim. In contrast, the presence of *Defenders* and their early action lead to abusers backing off with a lower impact on a victim. These results suggest that platforms could recalibrate engagement algorithms to limit the amplification of highly toxic content, particularly when metrics such as likes or retweets spike abnormally. Reducing the visibility of highly toxic posts would diminish reinforcement loops that escalate aggression. In addition, providing resources to bystanders to take a more active and prosocial role in digital environments and become active *Defenders* against abuse would lead to a significant reduction in bullying levels.

In the future, this model can be enhanced with sensitivity analyses to evaluate how small adjustments to exposure thresholds, reporting mechanisms, or content-ranking policies influence overall aggression stability. The insights provide a practical foundation for designing digital environments that better balance engagement and psychological safety by encouraging bystanders to become *Defenders*.

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